[METRIC]
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March 14, 1996
SUPERSEDING
MIL-C-10387J
10 April 1992

COMMERCIAL ITEM DESCRIPTION

COUPLINGS, CLAMP, PIPE; WITH BOLTS AND SYNTHETIC-RUBBER GASKETS FOR GROOVED-END PIPE

The General Services Administration has authorized the use of this commercial item description (CID) for all federal agencies.

1. SCOPE.

1.1 Scope. This CID covers clamp-type couplings, with bolts and gaskets, for grooved-end pipe, used with fuels.

2. CLASSIFICATION. Couplings (including gaskets) and gaskets for use with petroleum fuel shall be of the following classes and sizes (see 7.2 and 7.4):

2.1 Class (coupling or gasket). The coupling or gasket shall be of the following classes:

a. Aluminum (including gasket).
b. Malleable iron, ductile iron, or steel (including gasket).
c. Gasket only.

Beneficial comments, recommendations, additions, deletions clarifications, etc. and any other data which may improve this document should be sent to: U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/BLUE, Warren, MI 48397-5000.

AMSC N/A FSC 4730

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2.2 Size [outside diameter in millimeters (mm) (NPS designator)]. The coupling or gasket shall be of the following sizes:

26.7 (3/4)
33.4 (1)
42.2 (1-1/4)
48.3 (1-1/2)
60.3 (2)
73.0 (2-1/2)
88.9 (3)
101.6 (3-1/2)
114.3 (4)
141.3 (5)
168.3 (6)
219.1 (8)
273.0 (10)
323.8 (12)
355.6 (14)

3. SALIENT CHARACTERISTICS

3.1 Description. The coupling for joining two pieces of pipe or pipe fittings shall consist of a two-piece semicircular housing, a synthetic rubber gasket of pressure responsive design, and nuts and bolts, or other type of fasteners to secure the two halves together.

3.2 Materials. Unless otherwise specified herein, materials shall be in accordance with the manufacturer's specifications. The use of recovered metal made in compliance with regulatory requirements is acceptable providing all the requirements of the CID are met (see 4.1).

3.2.1 Class A - Aluminum housing. Aluminum permanent mold castings shall confirm to ASTM B108, alloy 356.0 (UNS A03560) in T6 temper. Aluminum sand castings shall conform to ASTM B26, alloy 356.0 (UNS A03560) in T6 temper. Aluminum forgings shall conform to ASTM B247 alloy number 6061 in T6 temper.

3.2.2 Class B - Malleable iron, ductile iron, and steel housing. Malleable iron castings shall conform to ASTM A47, grade 35018 or grade 32510. Ductile (nodular) iron castings shall conform to ASTM A536, grade 65-45-12. Steel castings shall conform to ASTM A27, grade 65-35.

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3.2.3 Material deterioration prevention and control. The couplings shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the couplings may be exposed.

3.2.3.1 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and subcomponents, and shall make information available upon request to the contracting officer or designated representative.

3.3 Coupling housing. Coupling housings shall be bolted over grooved-end pipe sections. The coupling shall provide an all-around grip on the grooved-end pipe with positive flexible locking action. The amount of flexibility on cut grooved pipe (see figure 1) shall be in accordance with table I. On roll grooved pipe, the deflection shall be half these values. The coupling housing shall be designed such that it does not pinch the gasket. Bolts for joining the coupling halves shall be zinc plated steel (see 3.5.1).

TABLE I. Coupling flexibility.

Outside Diameter in millimeters	Flexibility (Cut Grooved Pipe)	
	Minimum Degrees - Minutes	Maximum Degrees - Minutes
26.7 (3/4)	2°40′	7°0′
33.4 (1)	2°10′	5°45′
42.2 (1-1/4)	1°10′	4°35′
48.3 (1-1/2)	1°0′	4°0′
60.3 (2)	1°55′	3°15′
73.0 (2-1/2)	1°16′	2°45′
88.9 (3)	1°3′	2°15′
101.6 (3-1/2)	0°54′	2°42′
114.3 (4)	2°12′	3°25′
141.3 (5)	2°10′	2°25′
168.3 (6)	1°21′	2°25′
219.1 (8)	1°3′	1°55′
273.0 (10)	50′	1°35′
323.8 (12)	42′	1°25′
355.6 (14)	38′	1°15′

3.4 Gasket. The gasket shall be made from nitrile rubber and shall be compatible for use with fuels such as diesel fuels and jet fuels. Gaskets shall perform satisfactorily at all temperatures from minus (-) 29 degrees Celsius (°C) [-20 degrees Fahrenheit (°F)] to plus (+) 70°C (+158°F). The gasket shall be made from injection-molded or compression-molded synthetic rubber. Gasket

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size shall be identified with the outside diameter. Physical properties of the gasket shall meet or exceed the requirements shown in table II.

TABLE II. Physical requirements of the gasket.

Description of Properties	Requirements	ASTM Test Methods
Original Properties		
Tensile strength kilopascals (minimum)	10350	D412
Ultimate elongation (% minimum)	300	D412
Hardness, shore A durometer, points	60 + 15 - 10	D2240
After Oven Aging for 22 Hours at 70°C (158°F)		
Tensile retained (% minimum)	80	D573
Elongation retained (% minimum)	80	D573
Hardness change, shore A, points (maximum)	± 10	D2240
Compression set (% minimum)	45	D395 Using Method B
After immersion in ASTM Reference Fuel B for 70 hours at 23 ± 2°C		
Tensile retained (% minimum)	45	D412
Volume change (% maximum)	25	D471
Elongation retained (% minimum)	50	D471
Low temperature properties after extraction in ASTM Reference Fuel B for 70 hours at 23 ± 2°C, dried and cold conditioning at -31.7°C (-25°F) for seven days.		
Compression set (% maximum)	60	D1229
Torsional Stiffness Ratio (maximum)	10	D1053

3.5 Fasteners.

3.5.1 Bolts and nuts. The bolts shall conform to ASME B18.10, ASTM A183, grade 2 or ASTM A193, grade B7. Nuts shall conform to ASTM A194, grade 2H, heavy hex zinc type. Both bolts and nuts shall be zinc plated in accordance with ASTM B633, Fe/Zn-12.

3.5.2 Retaining pins. If retaining pins are used in lieu of bolts and nuts, pins shall be manufactured of ASTM A108, grade 1045, carbon steel, or ASTM A331, grade 4140, grade 4142, grade 4340, or grade 6150, pins shall be heat treated to Rockwell C hardness of 30 to 40. Pins shall have enough chamfer for ease of installations. The pins shall be zinc plated in accordance with ASTM B633, Fe/Zn-12. Retaining pins shall be captively (preferably with type) retained to the coupling.

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3.6 Pressure ratings. Ductile/malleable iron or steel couplings shall meet the working and test pressure ratings in table III on standard wall steel pipe. Aluminum couplings shall meet the working and test pressure ratings in table III on standard wall steel, aluminum, or stainless steel pipe. Couplings shall be tested to hydrostatic pressures as shown in table III for a period of not less than 5 minutes without evidence of leakage or permanent deformation.

TABLE III. Pressure ratings.

NPS designator	Outer Diameter mm (in)	Ductile Iron Coupling		Aluminum Coupling	
		Working pressure rating kiloPascals (kPa) [pounds per square inch (psi)]	Hydrostatic test pressure kPa (psi)	Working pressure kPa (psi)	Hydrostatic test pressure kPa (psi)
3/4	26.7 (1.050)	6900 (1000)	13800 (2000)	--	--
1	33.4 (1.315)	6900 (1000)	13800 (2000)	3450 (500)	5500 (800)
1 1/4	42.2 (1.660)	6900 (1000)	13800 (2000)	--	--
1 1/2	48.3 (1.900)	6900 (1000)	13800 (2000)	3450 (500)	5500 (800)
2	60.3 (2.375)	6900 (1000)	13800 (2000)	3450 (500)	5500 (800)
2 1/2	73.0 (2.875)	6900 (1000)	13800 (2000)	3450 (500)	5500 (800)
3	88.9 (3.500)	6900 (1000)	13800 (2000)	3450 (500)	5500 (800)
3 1/2	101.6 (4.000)	6900 (1000)	10350 (1500)	--	--
4	114.3 (4.500)	6900 (1000)	10350 (1500)	3450 (500)	5500 (800)
5	141.3 (5.563)	6900 (1000)	10350 (1500)	3450 (500)	5200 (750)
6	168.3 (6.625)	6900 (1000)	10350 (1500)	3450 (500)	5200 (750)
8	219.1 (8.625)	5500 (800)	8300 (1200)	2760 (400)	4140 (600)
10	273.0 (10.750)	5500 (800)	8300 (1200)	--	--
12	323.8 (12.750)	5500 (800)	6900 (1000)	--	--
14	355.6 (14.000)	2065 (300)	5500 (800)	--	--

3.7 Performance requirements. The coupling including the gasket shall be operable at all temperatures from -29°C (-20°F) to +70°C (+158°F), under working pressure from 508 mm (20 in) of mercury vacuum to 6900 kPa (1000 psi) (or rated working pressure of 8, 10, 12, and 14 nominal pipe size coupling) without any leakage or permanent deformation.

3.8 Treatment and painting. Unless otherwise specified (see 7.2), all exterior surfaces of couplings shall be cleaned, treated and painted in accordance with the manufacturer's standard commercial practice, using water-based acrylic emulsion alkyd enamel.

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3.9 Identification marking. Identification marking shall be permanent, legible and shall include as a minimum, the manufacturer's identification code (CAGE), the manufacturer's part number, the part or identification number (PIN), and the national stock number (NSN) (see 7.2).

4. REGULATORY REQUIREMENTS

4.1 Recovered material. The offeror/contractor is encouraged to use recovered metal to the maximum extent practicable in accordance with paragraph 23.403 of the Federal Acquisition Regulations (FAR).

5. QUALITY ASSURANCE PROVISIONS

5.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections (examinations and tests).

5.2 Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this CID and that the product conforms to the producer's own drawings, specifications, workmanship standards, and quality assurance practices. Items with known defects shall not be submitted for Government acceptance. The Government reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

6. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order (see 7.2).

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Non-Government documents. Copies of ASME B18.10 "Truck Bolts and Nuts" are available from the American Society of Mechanical Engineers (ASME), United Engineering Center, 345 East 47th Street, New York, NY 10017. Copies of ASTM B26 "Aluminum Alloy Sand Castings", ASTM A27 "Steel Castings, for General Application", ASTM A47 "Ferritic Malleable Iron Castings", ASTM A108 "Steel Bars, Carbon, Cold Finished, Standard Quality", ASTM B108 "Aluminum-Alloy Permanent Mold Castings", ASTM A183 "Carbon Steel Truck Bolts and Nuts", ASTM A193 "Alloy-Steel and Stainless Steel Bolting Materials for High-Temperature Service", ASTM A194 "Carbon and Steel Nuts for Bolts for High-Pressure and High-Temperature Service", ASTM A247 "Aluminum and Aluminum-Alloy Die Forgings, Hand Forgings, and Rolled Ring Forgings", ASTM A331 "Steel Bars, Alloy, Cold-Finished", ASTM A536 "Ductile Iron Castings", and ASTM B633 "Electrodeposited Coatings of Zinc on Iron and Steel". ASTM References are available from the American Society for Testing and

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Materials (ASTM), 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959. Copies of AWWA C606 "Grooved and Shouldered Joints" are available from the American Water Works Associations (AWWA), 6666 W. Quincy Avenue, Denver, CO 80235.

7.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this CID.
b. Issue of the Department of Defense Index of Specifications and Standards (DODISS) to be cited in the solicitation.
c. Class and size required.
d. When treatment and painting is other than as specified.
e. PIN number and quantity of bolts required.
f. Selection of applicable level of preservation, packaging and marking requirements.

7.3 Cross reference data. Couplings conforming to this CID are interchangeable/substitutable with couplings of the same part number conforming to MIL-C-10387J.

7.4 Part or identifying number (PIN). The PIN to be used for couplings acquired to this document are created as follows (see 7.2):

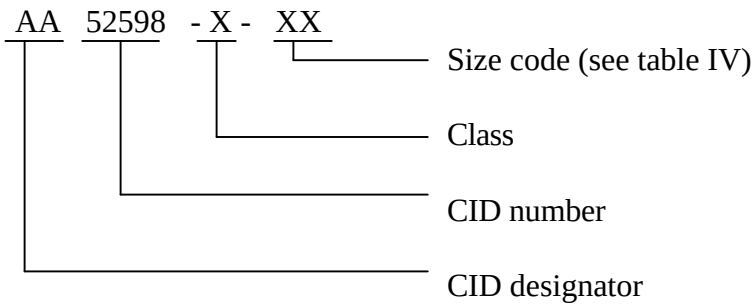

TABLE IV. Size code number.

NPS Designator	Size code number	NPS Designator	Size code number
3/4	01	4	09
1	02	5	10
1-1/4	03	6	11
1-1/2	04	8	12
2	05	10	13
2-1/2	06	12	14
3	07	14	15

7.5 Supersession data American Water Works Associations (AWWA) specification AWWA C606, supersedes the portion of MIL-C-10387H, dated 3 March 1982, for the couplings for use with water.

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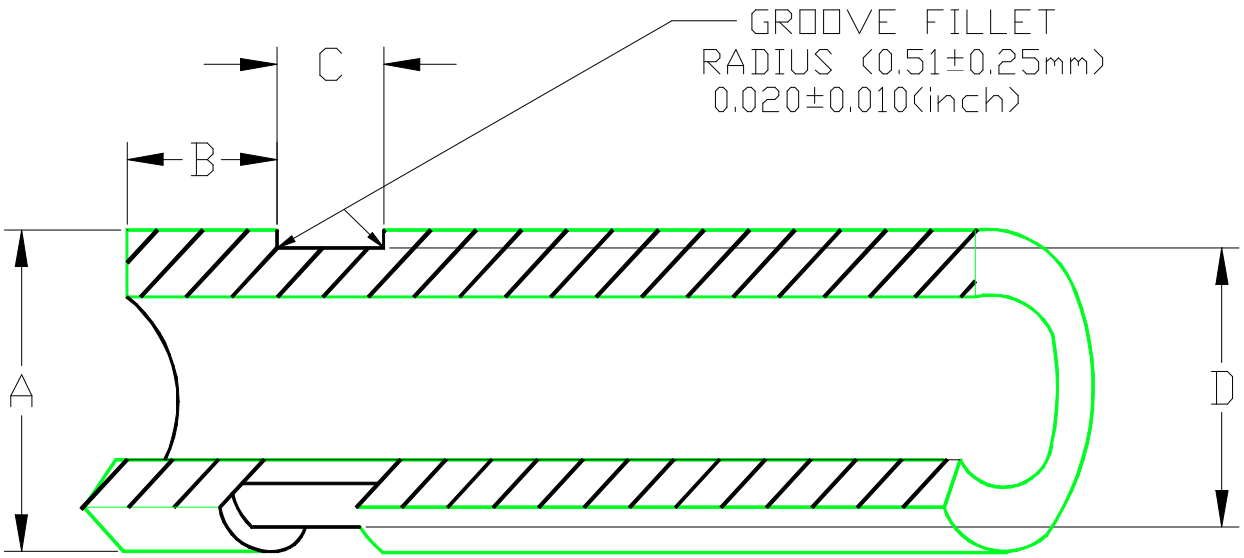
7.6 International standardization agreements. Certain provisions of this CID (for groove dimensions, see figure 1) are the subject of International Standardization Agreement NATO STANAG 3756. When amendment, revision, or cancellation of this CID is proposed which will modify the international agreement concerned, the preparing activity will take appropriate action through international standardization channels including departmental accommodations.



Groove Dimension (millimeters)				
NPS designator	Outside diameter A	Gasket seat B width	Groove width C	Groove diameter D 1/
3/4	26.7 ± 0.25	15.88 ± 0.76	7.95 ± 0.76	23.82 + 0.00 - 0.38
1	33.4 ± 0.33	15.88 ± 0.76	7.95 ± 0.76	30.23 + 0.00 - 0.38
1 1/4	42.2 ± 0.41	15.88 ± 0.76	7.95 ± 0.76	38.99 + 0.00 - 0.38
1 1/2	48.3 ± 0.48	15.88 ± 0.76	7.95 ± 0.76	45.09 + 0.00 - 0.38
2	60.3 ± 0.61	15.88 ± 0.76	7.95 ± 0.76	57.15 + 0.00 - 0.38
2 1/2	73.0 ± 0.74	15.88 ± 0.76	7.95 ± 0.76	69.09 + 0.00 - 0.45
3	88.9 + 0.89 - 0.79	15.88 ± 0.76	7.95 ± 0.76	84.94 + 0.00 - 0.45
3 1/2	101.6 + 1.02 - 0.79	15.88 ± 0.76	7.95 ± 0.76	97.38 + 0.00 - 0.51

FIGURE 1. Grooved pipe (millimeter).

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Groove Dimension (millimeters)				
NPS designator	Outside diameter A	Gasket seat B width	Groove width C	Groove diameter D 1/
4	114.3 + 1.14 - 0.79	15.88 ± 0.76	9.53 ± 0.76	110.08 + 0.00 - 0.51
5	141.3 + 1.42 - 0.79	15.88 ± 0.76	9.53 ± 0.76	137.03 + 0.00 - 0.56
6	168.3 + 1.60 - 0.79	15.88 ± 0.76	9.53 ± 0.76	163.96 + 0.00 - 0.56
8	219.1 + 1.60 - 0.79	19.05 ± 0.76	11.13 ± 0.76	214.40 + 0.00 - 0.64
10	273.01 + 1.60 - 0.79	19.05 ± 0.76	12.70 ± 0.76	268.27 + 0.00 - 0.69
12	323.9 + 1.60 - 0.79	19.05 ± 0.76	12.70 ± 0.76	318.29 + 0.00 - 0.76
14	355.6 + 1.60 - 0.79	23.83 ± 0.76	12.70 ± 0.76	350.04 + 0.00 - 0.76

1/ = See 7.6

FIGURE 1. Grooved pipe (millimeter) - Continued.

Groove Dimension (inches)				
NPS designator	Outside diameter A	Gasket Seat B width	Groove width C	Groove diameter D 1/
3/4	1.050 + .010	0.625 ± .03	0.313 ± .03	0.938 + 0.000 - 0.015
1	1.315 + .013	0.625 ± .03	0.313 ± .03	1.190 + 0.000 - 0.015
1 1/4	1.660 + .016	0.625 ± .03	0.313 ± .03	1.535 + 0.000 - 0.015
1 1/2	1.900 + .019	0.625 ± .03	0.313 ± .03	1.775 + 0.000 - 0.015
2	2.375 ± .024	0.625 ± .03	0.313 ± .03	2.250 + 0.000 - 0.015
2 1/2	2.875 ± .029	0.625 ± .03	0.313 ± .03	2.720 + 0.000 - 0.018

FIGURE 1. Grooved pipe (inches).

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Groove Dimension (inches)				
NPS designator	Outside diameter A	Groove seat B width	Groove width C	Groove diameter D 1/
3	3.500 + .035 - .031	0.625 ± .03	0.313 ± .03	3.344 + 0.000 - 0.018
3 1/2	4.000 + .040 - .031	0.625 ± .03	0.313 ± .03	3.834 + 0.000 - 0.020
4	4.500 + .045 - .031	0.625 ± .03	0.375 ± .03	4.334 + 0.000 - 0.020
5	5.563 + .056 - .031	0.625 ± .03	0.375 ± .03	5.395 + 0.000 - 0.022
6	6.625 + .063 - .031	0.625 ± .03	0.375 ± .03	6.455 + 0.000 - 0.022
8	8.625 + .063 - .031	0.750 ± .03	0.438 ± .03	8.441 + 0.000 - 0.025
10	10.750 + .063 - .031	0.750 ± .03	0.500 ± .03	10.562 + 0.000 - 0.027
12	12.750 + .063 - .031	0.750 ± .03	0.500 ± .03	12.531 + 0.000 - 0.030
14	14.000 + .063 - .031	0.938 ± .03	0.500 ± .03	13.781 + 0.000 - 0.030

1/ = See 7.6

FIGURE 1. Grooved pipe (inches) - Continued.

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - MC
Air Force - 99

Review Activities:
Army - CE1
Air Force - 82
DLA - CS

CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - AT

(Project 4730-0461)

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